



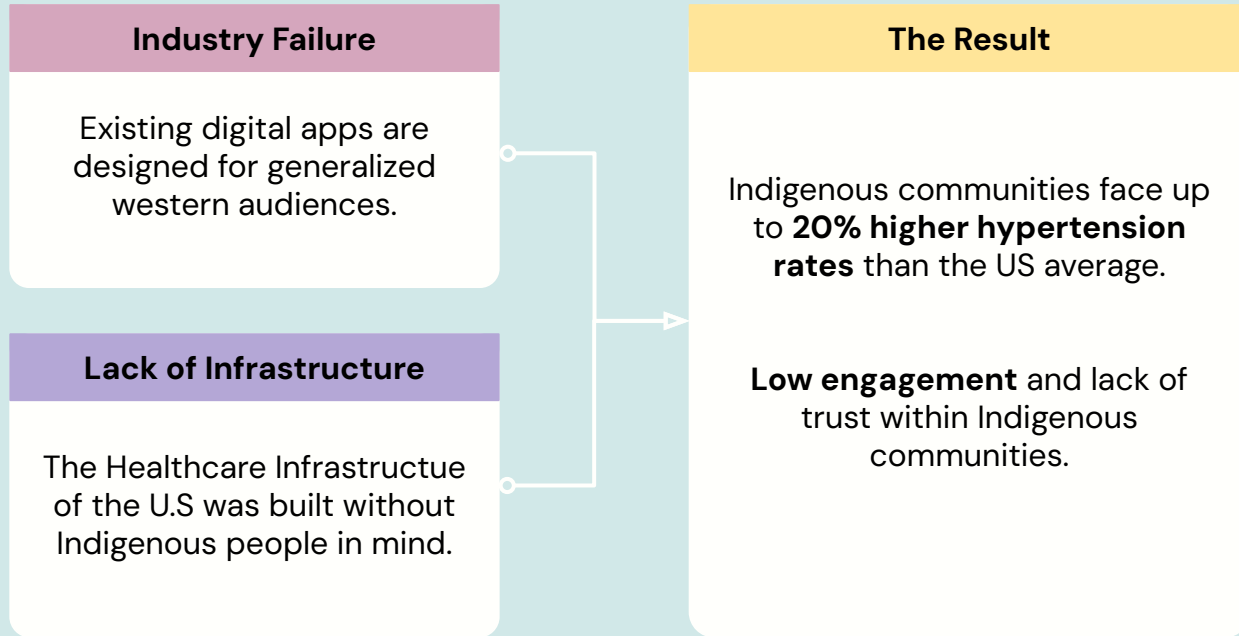
Team Heartline

CardioCare Quest: DR

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Rique Yazzie

Client: Dr. Jared Duval
Mentor: Morgan Nicholson

The Problem



Problem - Client



Dr. Jared Duval

Director of Playful Health Technology Lab

Studies human-computer interaction technologies.

Currently uses these technologies to remedy hypertension disparities in Indigenous populations.

Has identified an issue with the technology Twine, that being a steep learning curve for non-technical users.

Solution Overview

CardioCare Quest

Mobile health gaming environment for collecting hypertension data.

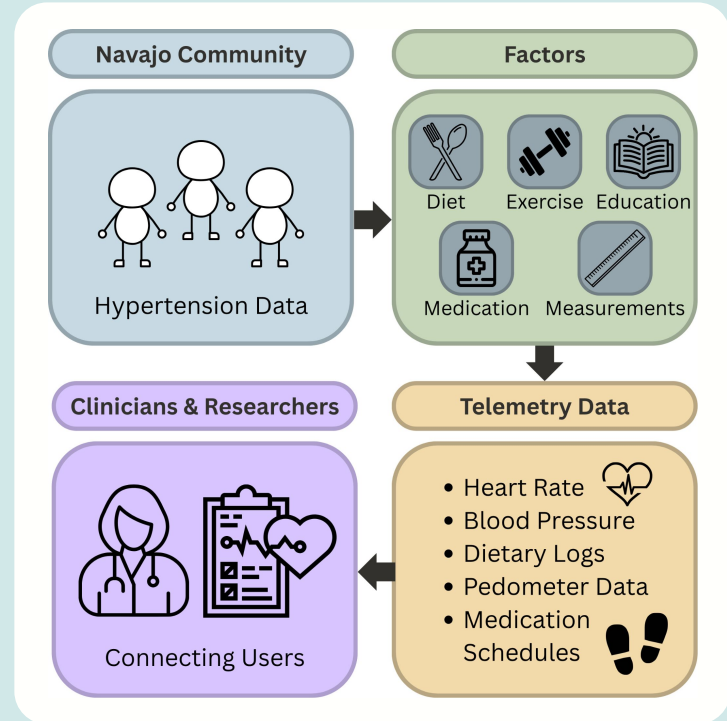
Exemplar Games

Five specialized Twine games designed as blueprints **for the community.**

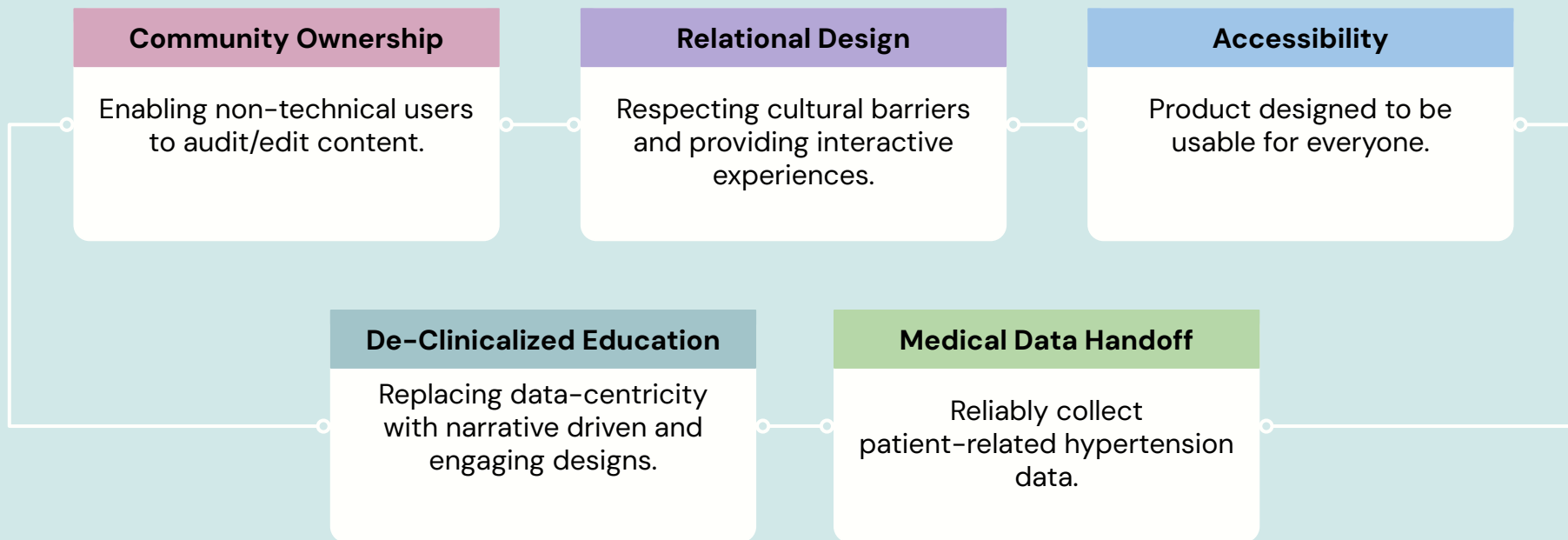


Unified App Shell

Flutter-based interface for navigating between games.



Key Requirements - Domain



Key Requirements - Overview

Where They Come From



Designs made for the community, by the community.

Methodology

- Compiled a comprehensive list of FR and NFR for each of the 5 games with respect to Domain and Environmental Requirements.
- Organized into levels:
 1. **User Level**
 2. **Game Level**
 3. **Data Handoff Level**
 4. **Global Standard**

Key Requirements - Dieting Factor

Level	Component	Functional Requirement	Non-Functional Requirement
High-Level	Relational Onboarding	Initializing a “chef” companion to guide tasks.	Feedback is delivered 100% accurately and by the companion.
Mid-Level	Interactive Tracking	Ingredient sorting for DASH diet compliance.	<100ms UI latency for any input.
Low-Level	Clinical Bridge	Packaging data into a JSON for the flutter shell.	100% of the provided metrics will be handed off successfully.
Global Standard	Accessibility	Game dialoge must be WCAG compliant.	Content remains functional with <=200% text resizing.

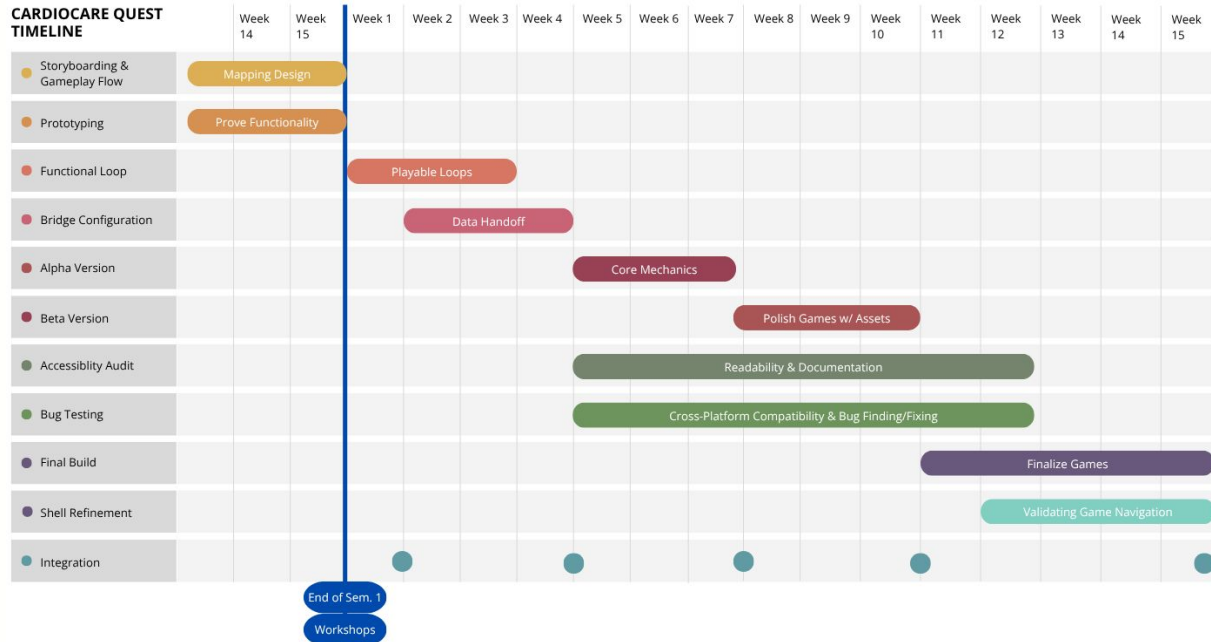
Risks & Feasibility



Severity	Risks	Feasibility
High	Accidental Data Exposure	Firestore Analysis
Moderate	Users being rewarded for false reporting	Clinical Tool Integration a. User Access to tools
Low	Moderating User Generated Content	Size and Future Scaling

Project Schedule

CARDIOCARE QUEST TIMELINE



Completed

- Requirements Acquisition
- Feasibility

In Progress

- Storyboarding
- Prototyping
- Workshopping

What's Next?

- First functional loops
- Bridge Configuration

Conclusion

The Problem

Indigenous communities face health and technological disparities

Internal Analysis

Through months of research and internal meetings, we have:

- Identified Requirements ranging from High to Low Level
- Conducted Risk Assessment
- Analyzed the Feasibility of Completion and Occurrence.
- Synthesized solid game concepts via workshops

Planned Solution

